

Graphs of Graphs of Graphs (G3): Capturing Transformations and Provenance in Knowledge Systems

Introduction

Every day, organizations take richly **structured data** – complete with context, lineage, and meaning – and flatten it into static documents like PDFs, slide decks, or reports. In doing so, they effectively create *snapshots* or “photos” of the data that are disconnected from their source materials ¹. This common practice introduces what some experts call **data entropy**: the loss of context and provenance that occurs when structured information is compressed for human consumption ². An insightful analogy likens it to “*unbaking a cake*”: a 50-page report might have begun as a row in a database or a sensor measurement, but once turned into a PDF, the original structure and metadata are stripped away ². Later, we spend enormous effort and money trying to reverse-engineer these documents – using tools like OCR, ETL pipelines, or even AI – to *guess* the structure and facts we ourselves destroyed in the flattening process ³. This cycle not only wastes resources, it also **breaks provenance, explainability, and determinism** in our data workflows ¹. In regulated or high-stakes environments, such loss of lineage undermines trust and hinders compliance, since one cannot easily verify how a piece of information was produced.

There is a growing recognition that we must “*stop fixing unstructured data and start preventing it*” ⁴. In other words, instead of relying on brittle after-the-fact reconstruction of context, we should capture knowledge in a **native, connected form from the start**. The approach advocated in this paper is to use **Graphs of Graphs of Graphs (G3)** – an emerging paradigm in knowledge management – to preserve **context** and **transformation lineage** throughout the data lifecycle ¹. By representing information as interconnected **knowledge graphs**, and explicitly modeling the steps and logic that link one graph to another, we can maintain a continuous chain of meaning from raw source data to final insights. This white paper introduces the G3 concept and explains how it addresses the provenance gap. We discuss why capturing intermediate transformations is critical, how G3 leverages ontologies and taxonomies to connect multi-layer graphs, and what benefits (and challenges) arise from this approach. The target audience is **knowledge graph practitioners** and data architects seeking to build systems with robust provenance and explainable insights.

The Problem of Lost Provenance in Static Artifacts

Conventional knowledge management pipelines tend to focus on *inputs* and *outputs* while neglecting the in-between steps. Consider a typical scenario: an analyst pulls data from databases, applies some transformations or analysis, and produces a report with conclusions. The **source data** might live in a structured repository (a relational DB or a base knowledge graph), but the **analysis logic** and reasoning often end up documented only in a human-readable form (narrative text, charts, PDF) rather than as structured, reusable knowledge. Once the final report is generated, it is essentially a dead-end snapshot – it tells *what* the analyst concluded, but not *why* or *how*. The rich context (evidence, intermediate calculations, assumptions) is locked in prose or lost entirely. As Dinis Cruz observes, we ironically “*take*

structured data, evidence and reference materials, and then create these 'photos' of it that are static and not connected to the source", breaking the provenance and explainability of the information ¹.

This loss of lineage is not just an academic concern; it has real consequences. Without a **decision trace** or transformation record, one cannot easily **trust** the output or verify its correctness. In enterprise settings, this means audits become arduous and uncertainties abound – was a particular figure in the report derived from accurate data? What filters or assumptions were applied? Lacking structured provenance, we might have to retrace steps manually from scratch. In AI and machine learning contexts, models trained on these static artifacts lack awareness of how facts were derived, increasing the risk of errors or hallucinations. *There is no such thing as "naturally unstructured" data* in the first place – virtually all unstructured reports originated from structured inputs ². The unstructured state is an **artificial artifact of our documentation process**, and it entails throwing away useful information (lineage, context, semantics) that later we attempt to glue back. Figure 1 illustrates this problematic pattern of context loss:

[Source Data (Structured)] --X--> [Flattened Document (PDF/Text)]
(context & lineage lost)

Figure 1: Traditional workflow where structured source data is flattened into a static document, losing the context, metadata, and provenance (X). Later attempts to recover structure from the document (e.g., via parsing or AI) are costly and imperfect.

To address this, we need to change how we represent and share knowledge. Rather than static documents, **we require a living representation that carries its provenance along**. This is where knowledge graphs come in – and specifically, an evolution of knowledge graphs that can handle not just static facts, but also the *dynamics* of reasoning and transformation.

From Knowledge Graphs to Context Graphs: Capturing Transformations

Basic **Knowledge Graphs (KGs)** model factual knowledge as a network of entities and relationships – a *snapshot of what is true* in a domain ⁵. For example, a simple KG might state that *Customer A purchased Product B*, linking nodes for *Customer A* and *Product B* with a *purchased* relationship. Traditional KGs, however, are largely *static and state-oriented*: they excel at representing **nouns** (entities) and their connections, but not the **verbs** – the actions, events, or decisions that cause knowledge to change ⁶. In our running example, the KG alone might not capture *why* a certain refund was issued or *how* a particular insight was derived from raw data.

This gap has led to the concept of **Context Graphs (CGs)**, which build upon KGs by adding two critical dimensions: **time** and **decision lineage** ⁵. In essence, *a context graph is a knowledge graph plus a history of events (what happened when) and the rationale behind decisions*. Andreas Blumauer succinctly describes a Context Graph as *"effectively a Knowledge Graph + Time + Decision Lineage."* ⁵ In practice, this means that in addition to static facts, the graph captures *why those facts came to be or changed*. Context graphs treat **actions and inferences as first-class citizens** of the data model. For example, a context graph could record that *"Customer A was given a refund on 2025-12-01 because Policy X was overridden due to a Service Outage event"*, linking the event, the decision (override approval), the actors (perhaps a manager or automated system), and the policy involved ⁷ ⁸. This transforms a pure data state graph into a **story** of how that state came about.

Translating this to the realm of data analysis and knowledge capture, every time we derive new information or make an assumption, we can model it as an **event or transformation in a graph**. Think of each analytical step as a function: it takes some input data and produces some output data. Rather than only recording the final output, a context-aware knowledge graph will also record the *function applied* (the method, algorithm, or reasoning step) and link it to its inputs and outputs. In W3C's Provenance (PROV) model, this corresponds to an **Activity** (the transformation) that *used* certain source **Entities** and *generated* new result Entities ⁹. Provenance standards like PROV formally distinguish these elements – entities, activities, and agents – precisely so that we can reconstruct how a piece of data was produced ⁹. Notably, PROV-O even allows for a concept of **bundles** (or nested provenance) which means one can encapsulate an entire provenance trail as an entity in a higher-level provenance graph ⁹. This is essentially a “graph-of-graphs” approach to provenance: it enables provenance of provenance, or meta-provenance, to be tracked.

Modeling a Chain of Transformations (Ascii Diagram)

To make this concrete, imagine an analysis process with two steps: first, raw data is cleaned and aggregated; second, the aggregated results are interpreted to produce a final report. We can represent the knowledge at each stage as a graph, and the analysis steps as transformation nodes connecting them. Figure 2 shows this chain in a simplified form:

```
[ Raw Data Graph ] --(Step 1: Aggregate)--> [ Aggregated Graph ] --(Step 2: Interpret)--> [ Report Graph ]
```

Figure 2: Capturing transformation lineage in a graph-of-graphs. Each rectangle represents a knowledge graph at a certain stage. Each parenthesized label represents a transformation function or analysis step that consumes the prior graph and produces the next. Rather than discarding intermediate outputs, this approach preserves the Aggregated Graph and links it back to source data and forward to final report.

In this model, the **Raw Data Graph** might contain factual observations or records (for instance, transaction logs). The **Aggregated Graph** could contain derived statistics or summaries (e.g. sales totals, averages), generated by an *Aggregate* function that is documented in the system. The **Report Graph** then contains conclusions or recommendations which are generated by an *Interpret* step that took the aggregates as input. Each arrow is not a simple data flow arrow but an **explicitly documented transformation**. We know *which function or analysis* produced the Aggregated Graph from raw data, and which one produced the Report from aggregates. Crucially, we could attach metadata to these transformation steps – who performed them (analyst or algorithm), when they occurred, what assumptions or parameters were used – all of which builds a **decision lineage**. This end-to-end pipeline forms a **context graph** that the organization can query or traverse to answer questions like “Why does the final report say X?” or “Which source data contributed to this conclusion?” The answer would be discovered by following the links through the transformation nodes back to specific raw data points, each hop explaining part of the “why.”

Graphs of Graphs of Graphs (G3): A Multi-Layer Knowledge Framework

Capturing transformations as above already elevates a knowledge graph into a context-rich medium. **Graphs of Graphs of Graphs**, or **G3**, takes this a step further by organizing knowledge into **multiple interconnected graph layers or domains**. The core idea of G3 is that instead of having one monolithic graph trying to represent everything, we maintain **modular graphs that each describe a particular**

layer, aspect, or context, and then connect these graphs into a larger **graph-of-graphs structure** ¹⁰ . In a G3 system, *each graph can itself be treated as a node or component of a higher-level graph*, enabling a flexible “zoom in and out” between levels of detail.

Real-world knowledge is naturally multi-layered. Consider the following examples from different domains, which underscore that complex systems are effectively “*graphs of graphs of graphs*” in practice

¹¹ :

- **Software Engineering:** A compiler has an abstract syntax tree (a graph of code structure), a control-flow graph (for program execution paths), and an optimization graph or trace. These graphs interrelate (e.g., nodes in the control-flow graph link to syntax tree nodes for the corresponding code) ¹¹ .
- **Databases:** A database system manages a schema graph (tables and relations), query plan graphs (for each SQL query’s execution plan), and possibly a dependency graph of transactions or data lineage. Each query’s plan could be seen as a subgraph linked to the schema it operates on ¹² .
- **AI Systems:** An AI agent might maintain a knowledge graph as its world model, another graph for its internal goals or intentions, and a conversation or interaction graph as it engages with users. These layers feed into each other; e.g., an interaction event updates the world model graph ¹² .
- **Enterprise Knowledge:** An organization could have separate knowledge graphs for different domains – e.g., a customer graph, a product graph, a policy/regulation graph – plus event graphs capturing transactions or decisions. G3 would allow each departmental or domain graph to remain distinct yet **semantically interlinked** at the higher level.

In a G3 architecture, **ontology management** becomes critical. We often deal with **ontologies of ontologies** and **taxonomies of taxonomies**, meaning each domain or layer may have its own schema and classification system, and these themselves need to be connected or mapped ¹³ . There isn’t a single “Uber” ontology that covers everything ¹³ . Instead, G3 embraces plurality: multiple ontologies coexist, and we establish bridges or alignment between them. *The power of G3 lies in having ways to communicate and connect multiple graphs, domains, and ultimately different points of view* ¹³ . For example, one graph’s notion of a “Customer” might need to link to another graph’s notion of a “Person” or “Client” – they might not use identical definitions, but through an ontology mapping, the system knows these concepts relate. G3 thus often involves **meta-ontologies** or upper ontologies that define general concepts used to glue domain-specific ontologies together. Likewise, a taxonomy (classification scheme) in one context can be related to a taxonomy in another through cross-links, essentially creating a larger taxonomy-of-taxonomies.

Multi-Layer Graph Connections (Ascii Diagram)

Figure 3 illustrates an abstract G3 arrangement with two layers of graphs. In the higher-level Graph (Layer 1) we see two nodes, A and B, connected by a relationship. In the lower layer (Layer 2), we zoom into node A to reveal that A is itself a graph with internal nodes (A1, A2, A3 and their relationships):

Layer 1 Graph (Overview):

```
[Node A] -- relatedTo --> [Node B]
```

Layer 2 Graph (Detail of A):

```
Node A1 -- connects --> Node A2
```

```
Node A2 -- connects --> Node A3
... (etc.)
```

Figure 3: Graphs within graphs. In Layer 1, Node A and Node B are high-level conceptual nodes (which could represent complex entities or entire subsystems) with a relationship. In Layer 2, we see that Node A is not atomic – it expands into its own subgraph (A1, A2, A3 ...). The relationship in Layer 1 between A and B could be supported or explained by numerous relationships in the detailed layer (not shown for brevity). A G3 system lets us navigate from the abstract layer down into the granular layer and back up.

In this toy example, imagine *Node A* was “Supply Chain” and *Node B* was “Market Risk” in a corporate knowledge graph. The Layer 2 graph for *Node A* might contain nodes for individual suppliers, warehouses, shipments (and their relationships indicating the supply chain structure). The high-level link *A relatedTo B* could represent the fact that supply chain issues influence market risk. Thanks to the G3 structure, an analyst could click on “Supply Chain” (Node A) and dive into the detailed supply chain graph to investigate which specific elements (say, a delayed shipment node A2) contribute to a risk that propagates to the market. Conversely, from a low-level event (like A2 = “Shipment #123 delayed by 5 days”), one could traverse back up to see that it rolls up into the broader category of “Supply Chain disruptions” (Node A) which is linked to Market Risk (Node B) at the top level. This **zoom-in/zoom-out capability** is a hallmark of G3. It enables both **bird’s-eye view** and **microscopic view** using the same interconnected dataset.

Provenance and Ontology: Foundations of G3

To successfully implement a G3 approach, two foundational pillars are needed: **Provenance capture** and **semantic interoperability**. Provenance capture ensures we record *how* each piece of data or knowledge came to be; semantic interoperability ensures that when we connect multiple graphs, we understand each other’s terms and context.

Provenance capture in G3 means every graph or dataset carries with it a description of its origins and the transformations it underwent. This is where standards like **W3C PROV** become incredibly useful. PROV provides an ontology to represent provenance information in a graph form – detailing entities, activities, and agents involved in producing data ⁹. By pairing a knowledge graph with PROV annotations, we can get “*verifiable decision trails*”, as one CIO article noted ¹⁴. In other words, any node or edge in our knowledge graph can point to a subgraph explaining its derivation (which source nodes, which process, at what time, and by whom). This is essential for **explainability** and **auditability**. A knowledge graph that can answer not just factual queries (“What is the revenue?”) but also provenance queries (“How was this revenue figure calculated and from which raw data?”) becomes a powerful tool for trust. In fact, provenance itself can be stored as a knowledge graph – often called a **provenance graph** – and connected to the primary knowledge graph. Thanks to the PROV model (and similar frameworks), this provenance graph can itself have provenance (i.e., you can have meta-provenance about how the provenance was assembled), aligning perfectly with the G3 vision of graphs connecting to graphs.

Semantic interoperability is the other side of the coin. G3 systems link multiple graphs that might have been created in different formats, by different teams, or for different purposes. Without a common semantic layer, such connections would be ambiguous or lossy. This is why **ontologies and taxonomies** play a central role. An ontology formally defines the types of entities, attributes, and relationships in a domain (for example, defining what “Customer” means and how it relates to “Order” or “Product” in a retail domain). In a multi-graph ecosystem, we likely have multiple ontologies – some domains might overlap, some might be hierarchical extensions of others. G3 does not enforce a single

mega-ontology; rather, it encourages linking and mapping ontologies. For instance, a higher-level *upper ontology* (perhaps based on an industry standard or an abstract framework like ISO 21838-2) can provide common ground, while each sub-graph uses a domain-specific ontology aligned to that upper model ¹⁵ ¹⁴. Additionally, techniques like **ontology alignment** or **schema mapping** are employed to translate or connect concepts between graphs. We might declare that *Graph A's "Client" = Graph B's "Customer"*, or that *Graph C's "Incident" is a subclass of Graph D's "Event"*. By doing so, queries and inferences can flow across graph boundaries. When an analyst asks a question that spans multiple sub-graphs, the system can reconcile the terminology via these ontology mappings and return an integrated answer.

Practically, modern knowledge graph technology provides tools for this semantic layering. We have **named graphs** and **RDF-star** (an extension of RDF) that allow attaching context (like timestamps, confidence, or provenance) to specific triples in a graph ¹⁶. Named graphs can represent each sub-graph as a identifiable unit within a triple store, making it easier to manage graph-of-graphs structures. For provenance, RDF-star enables annotations on relationships (for example, marking a relationship with the time it was asserted or the source). These capabilities align well with context graphs: as Blumauer notes, a context graph often *"adds a meta-layer to the knowledge graph, using RDF-Star or Named Graphs to attach timestamps, provenance, and confidence scores to links"* ¹⁶. In summary, by using semantic web standards and ontology frameworks, we ensure that our G3 implementation isn't a patchwork of isolated silos, but a **coherent semantic network** where each piece knows how to talk to the others.

Benefits of the G3 Approach

Adopting a Graphs-of-Graphs-of-Graphs approach yields numerous benefits for organizations aiming to leverage their data intelligently:

- **End-to-End Provenance & Trust:** G3 provides *full traceability* of knowledge from origin to insight. Every node and edge in a final graph can be traced back through intermediate graphs to raw data sources, along with the transformations applied ¹⁴. This builds trust with stakeholders (e.g. regulators, scientists, executives) because claims and decisions are backed by an auditable trail. It's compliance-ready evidence of "how you know what you know." If a question arises about a particular decision, one can traverse the graph-of-graphs and inspect each step's justification, rather than accepting the final report on faith.
- **Explainability and Decision Lineage:** Because G3 captures **why** things happened (not just what happened), it naturally supports explainable AI and human decision-making. As noted, context graphs act as a "system of agency" recording why judgments were made ⁷. For instance, an AI assistant or an analyst can refer to the context graph to explain *"We recommended action Z because factors X and Y were present and policy Q was met"*. The decision lineage is explicit. In scenarios like healthcare or finance, where explanations are as critical as the decisions themselves, this is a game-changer.
- **Modularity and Reusability:** Instead of one giant model, knowledge is organized into **modular graphs** that correspond to domains or layers. This makes models more maintainable and extensible. Teams can work on their domain-specific graph (e.g., a product catalog graph, or a legal compliance graph) and later link it with others without having to merge everything into one schema. It also encourages reuse of existing knowledge – e.g., linking to a published ontology or an industry taxonomy – rather than reinventing the wheel. G3's *"ontologies of ontologies"* philosophy means you can plug one ontology into another gracefully ¹³. If a new data source or

domain comes along, you add another graph and connect it, instead of rebuilding the world model from scratch.

- **Multi-perspective Views:** Different stakeholders often need to view data at different levels of granularity or from different perspectives. G3 inherently supports **multi-view visualization**. With layered graphs, a high-level executive dashboard and a low-level technical diagnostic view can be derived from the same interconnected knowledge base. This eliminates the consistency problems that occur when separate teams maintain separate views that drift over time. In a G3 system, the views are slices of the same truth. As we saw in the Layered Graph example, one can pivot the graph to focus on a subgraph of interest without losing the context that connects it to broader concerns.
- **Prevention of Data Entropy:** By capturing information in structured form throughout the lifecycle, we avoid the costly loss-and-recreation of context. The phrase “*better capture (at the source) instead of better RAG (retrieval after the fact)*” encapsulates this benefit ⁴. In practical terms, that means less time cleaning and wrangling data for each new analysis, and more time using it. Knowledge remains in a **machine-readable, queryable form** at all times. Integration with AI becomes smoother as well – LLMs or other AI agents can be fed with context graphs that contain not just raw facts but also the narrative of those facts, reducing guesswork.
- **Determinism and Reproducibility:** When each transformation is logged and linked, analyses become reproducible. Another analyst (or an automated system) can follow the graph to rerun or verify each step. If something looks suspicious in the final output, one can pinpoint where it entered the pipeline. This is crucial for scientific research, data science, and any data-driven business process. It moves us closer to **self-documenting analyses** – the documentation isn’t a separate afterthought but an integral part of the data model.

To illustrate the synergy of these benefits, consider an enterprise that implemented a G3-based “knowledge OS.” Such an enterprise pairs its knowledge graphs with a provenance graph and context layers. When the CEO asks a question, the answer not only comes back as a factual result, but also carries an explanation graph. Internally, their AI systems perform reasoning by traversing these context graphs, ensuring that even automated decisions respect the business rules and past precedents recorded. The company can onboard new data sources rapidly by spinning up new graphs and mapping them to the existing ontology network, accelerating integration. And if an auditor comes knocking, every number in every report can be clicked to reveal its supporting data and the chain of reasoning behind it. This is the promise of G3 in practice.

Challenges and Considerations

While the G3 approach is powerful, it is not without challenges. Implementing a graphs-of-graphs system requires careful thought in the following areas:

- **Complexity Management:** Introducing multiple layers of graphs and extensive provenance tracking can increase complexity. There is a risk of **information overload** if the system is not designed with clear abstraction boundaries. Users might be overwhelmed if every data point expands into a huge subgraph. Good UX and tooling are needed to let users seamlessly navigate complexity – e.g., toggling levels of detail, filtering relevant lineage for a given query, etc. The key is to expose just enough of the context graph to answer the question at hand, without drowning in details.

- **Performance and Scalability:** Storing “graphs of graphs” and querying across them can be demanding. Each query might traverse multiple layers or follow long provenance chains. Graph databases and triple stores need to handle such traversals efficiently. Caching and index optimization (for example, precomputing some lineage links) may be necessary for interactive performance. Moreover, writing data with provenance means more triples per fact (to encode the source, time, agent, etc.), so storage and transaction volumes will grow. However, advances in graph database technology and the adoption of scalable knowledge graph frameworks (many of which are cloud-native) have begun to make this feasible at enterprise scale.
- **Standardization vs. Flexibility:** There’s a tension between using **standard ontologies/provenance models** and the need to adapt to specific domain requirements. PROV-O is a great foundation, but some domains might require custom extensions (e.g., special activity types or domain-specific provenance attributes). Similarly, aligning ontologies across disparate graphs can be non-trivial – it may involve negotiation between teams or the creation of new “bridge” ontologies. Governance processes are needed to maintain semantic consistency without stifling the agility of individual graph development. In practice, a balance must be struck between **global standards** (to enable integration) and **local autonomy** (to allow innovation in representing domain knowledge).
- **Cultural and Workflow Changes:** Perhaps the biggest barrier is not technical but human. Adopting G3 means analysts, engineers, and domain experts need to change how they work. Instead of jotting down conclusions in a Word document, they would contribute to a knowledge graph or annotate their findings with provenance metadata. This requires training and a mindset shift towards treating knowledge as code/data. Organizations may need to incentivize maintaining the graphs and provide easy-to-use interfaces so that adding to the context graph is as natural as writing a report. Success stories often involve gradually introducing these practices – for example, starting by capturing just critical decisions as provenance, or by integrating graph updates into existing review processes.

Despite these challenges, the trajectory of data management is clearly bending towards more context, more provenance, and more connectivity. The concept of G3 is resonating across domains, from cybersecurity threat modeling to enterprise knowledge integration, because the pain points of siloed and context-stripped information are universally felt. Notably, research efforts are underway to formally support nested graph structures (for instance, *WARP graphs*, which provide mathematical foundations for graphs containing subgraphs ¹⁷ ¹⁸). These advances suggest that tooling will catch up with the theoretical possibilities, making G3 more accessible and efficient over time.

Conclusion

As organizations grapple with ever-growing volumes of data and the need for intelligent, transparent decision-making, the limitations of static, provenance-poor artifacts have become stark. **Graphs of Graphs of Graphs (G3)**, underpinned by rich provenance capture and semantic linkages, offers a path forward. By embracing context graphs and layered knowledge representations, we stop discarding context and start **baking explainability and lineage in by design**. The result is knowledge systems that are not only machine-friendly but also human-trustworthy – where every insight can be traced back to its roots, and multiple viewpoints can coexist and inform each other within a unified framework.

In summary, G3 is about treating knowledge as a living, interconnected network rather than a series of one-off reports. It means capturing the *process* along with the *data*, linking **the “what” with the “why” and “how”** at every step. This approach addresses the “original sin” of data management – the artificial

separation of data from its context ¹⁹ – and moves us toward a future where our information systems have memory and understanding of their own evolution. For knowledge graph practitioners, adopting even elements of this paradigm (be it adding PROV metadata to your graphs, or integrating multiple domain ontologies) can significantly enhance the value and longevity of your data. The technology, standards, and methodologies are aligning to support G3: from PROV-O and RDF-star to graph databases that handle meta-graphs. The remaining step is organizational will to rethink data practices.

The writing on the wall is clear: **the future of enterprise AI and data analytics isn't about bigger black-box models operating on shredded context – it's about better capture and connected context** ³. Graphs of graphs of graphs provide a blueprint for that future, one where knowledge is preserved in full color rather than flattened to black-and-white. By capturing transformations and context as first-class data, we ensure that our knowledge systems remain **explainable, interoperable, and resilient** in the face of change. It's an evolution of knowledge representation that promises to unlock deeper insights while retaining the trust of those who rely on them. The task now is to implement these ideas in our tools and workflows, building the provenance-rich, context-aware knowledge infrastructure for the coming decades.

References: This white paper builds upon concepts from the W3C PROV standard for data provenance ⁹, recent discussions on context graphs vs. knowledge graphs ⁵ ²⁰, and thought leadership by Dinis Cruz and others on the “graphs of graphs” paradigm ¹ ¹³. The inefficiencies of current unstructured data practices and the need for better data capture were highlighted by industry voices ¹⁹. Early implementations of graph-of-graphs ideas, such as using linked sub-graphs in threat modeling ¹⁰ and formal models like WARP for nested graphs ¹¹, demonstrate the feasibility and value of G3 in practice. Together, these sources underscore a converging trend: moving beyond flat knowledge representations towards deeply linked, multi-layered graph ecosystems that mirror the complexity of the real world while preserving the context that makes data truly meaningful.

¹ ² ³ ⁴ ¹⁹ Addressing Provenance and Explainability in Data Representation | Dinis Cruz posted on the topic | LinkedIn

https://www.linkedin.com/posts/diniscruz_this-is-absolutely-spot-on-and-it-is-crazy-activity-7420815037105172480-huOT

⁵ ⁶ ⁷ ⁸ ¹⁶ ²⁰ How Do Context Graphs and Knowledge Graphs Differ From Each Other?

<https://www.linkedin.com/pulse/how-do-context-graphs-knowledge-differ-from-each-other-blumauer-wkfcf>

⁹ PROV-DM: The PROV Data Model

<https://www.w3.org/TR/prov-dm/>

¹⁰ graphs-of-graphs-of-graphs-g3-in-threat-modeling.md

<https://github.com/DinisCruz/docs.diniscruz.ai/blob/fa051954e6c82dbbabcdced29880d613cc095bf5/docs/2025/05/30/graphs-of-graphs-of-graphs-g3-in-threat-modeling.md>

¹¹ ¹² ¹⁷ ¹⁸ THEORY.md

<https://github.com/flyingrobots/echo/blob/19f32367a10ccd56a633eaf51d97a53df57ec50f/docs/THEORY.md>

¹³ This is exactly why my focus is on G3 (Graphs of Graphs of Graphs), where we have ontologies of Ontologies and Taxonomies of Taxonomies | Dinis Cruz

https://www.linkedin.com/posts/diniscruz_this-is-exactly-why-my-focus-is-on-g3-graphs-activity-7336340555905318914-DiBs

¹⁴ ¹⁵ Enterprise AI and the semantic core: Building an OS for knowledge | CIO

<https://www.cio.com/article/4122243/enterprise-ai-and-the-semantic-core-building-an-os-for-knowledge.html>